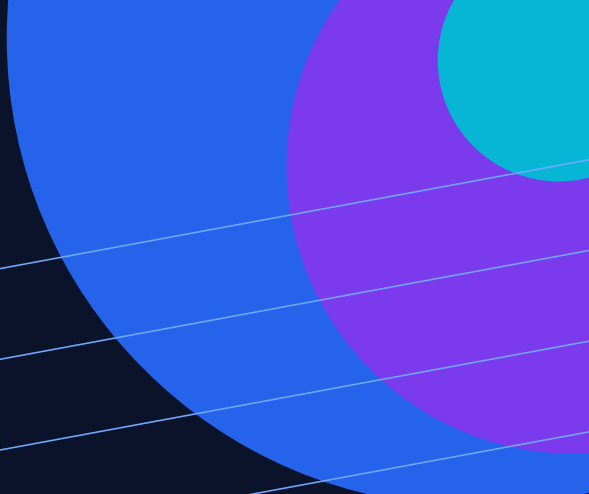




THREAT RESEARCH / DETECTION ENGINEERING

ClickLock macOS Kill-Loop

Repeated termination of core GUI processes to coerce credential entry

REPORT	STATUS	CONFIDENCE
Detection dossier	Production candidate	Medium

Prepared for defensive research and environment-specific validation. Source facts, analytical interpretation, assumptions and proposed detection logic are intentionally separated.

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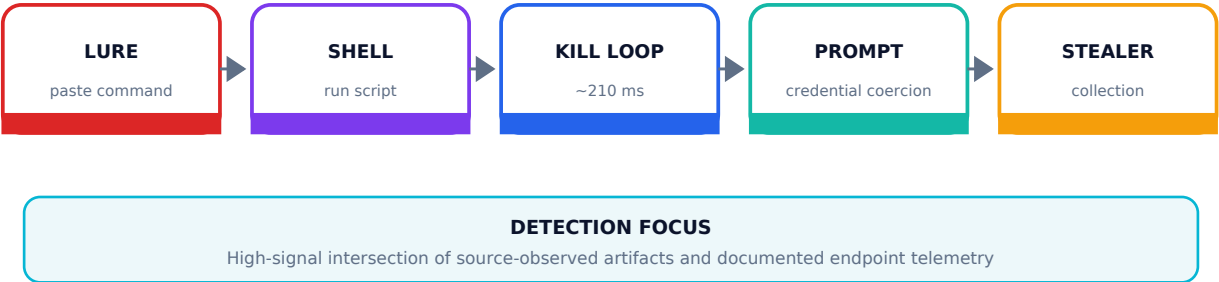
Executive Assessment

WHY THIS MATTERS

ClickLock repeatedly terminates visible macOS processes at very short intervals while presenting credential prompts. A one-minute burst of pkill/killall targeting core GUI components may provide a compact behavioral signal, subject to validation of short-lived process capture.

Dimension	Assessment	Operational Implication
Observed behavior	Source-analyzed modules repeatedly terminate applications and suppress NotificationCenter; persistence is created through LaunchAgents.	Frequency and target selection are more discriminating than the presence of pkill or killall alone.
Detection posture	Production candidate	The threshold is a conservative implementation assumption and must be baselined before alerting.
Telemetry	Microsoft Defender XDR DeviceProcessEvents on macOS	Provides process name, command line, user, device and parent process context.
Primary limitation	Short-lived process telemetry may be sampled or incomplete.	A false negative is possible even when the kill-loop occurs.

Attack and observation flow



Evidence Model

Evidence class	Statement	Use in the package
Source fact	Group-IB reports repeated process termination approximately every 210 ms and prolonged NotificationCenter suppression.	Directly supports the selected behavior or artifact.
Documented telemetry	DeviceProcessEvents documents Timestamp, FileName, ProcessCommandLine, account, device and initiating-process fields.	Supports the table, fields and investigation pivots.
Analytical interpretation	Twenty matching termination attempts in one minute is conservative relative to the reported loop but is not source-derived as an alert threshold.	Defines why the intersection may be distinguishable.
Unvalidated assumption	MDE captures enough brief pkill/killall executions and their target arguments to cross the threshold.	Must be measured before deployment or promotion.

EVIDENCE BOUNDARY

The query does not prove credential theft, distinguish the original lure, or detect every ClickLock module. It identifies a high-rate process-termination behavior compatible with the report.

Telemetry contract

Component	Requirement	Failure condition
Data source	Microsoft Defender XDR DeviceProcessEvents on macOS	macOS endpoints are not onboarded to MDE.
Core fields	Timestamp, DeviceId, DeviceName, FileName, ProcessCommandLine, AccountName, InitiatingProcessFileName, ReportId	Target process absent from command line or short executions sampled out.
Time context	One-minute aggregation; baseline across at least 30 days.	Aggregation granularity or ingestion loss prevents the burst from forming.

Detection Logic

HYPOTHESIS

A credential-coercion kill-loop should produce a dense burst of pkill/killall commands targeting Finder, Dock, SystemUIServer or NotificationCenter.

```
DeviceProcessEvents
| where FileName in~ ("pkill", "killall")
| where ProcessCommandLine has_any (
    "Finder", "Dock", "SystemUIServer", "NotificationCenter")
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp),
    TerminationAttempts=count(),
    Commands=make_set(ProcessCommandLine, 20),
    Parents=make_set(InitiatingProcessFileName, 10)
    by DeviceId, DeviceName, AccountName, bin(Timestamp, 1m)
| where TerminationAttempts >= 20
```

Condition	Purpose	Durability
pkill / killall	Selects macOS process-termination utilities.	High
GUI targets	Restricts to source-relevant visible system components.	Medium
>=20 per minute	Approximates a rapid loop while limiting isolated administration.	Low until baselined

Analyst Triage

Priority	Question	Evidence to collect
1	Which script or parent generated the burst?	Parent image, command line, script path and signer.
2	Were password prompts or LaunchAgents observed?	Unified logs, file events under LaunchAgents, osascript activity.
3	Was data collection initiated?	Archive creation, browser/password-manager access, Telegram traffic.
4	Is the device managed for kiosk or GUI testing?	MDM job, software owner and change record.

False-positive model

Scenario	Why it may match	Recommended handling
MDM remediation	Automation may restart GUI components.	Bind any exception to a verified script hash/path and management parent.
Kiosk or UI testing	Test frameworks may repeatedly reset applications.	Validate device role, schedule and owned automation.
Recovery tooling	Support scripts may terminate Finder or Dock.	Require documented operator and ticket context.

NO PREMATURE ALLOWLISTING

Do not allowlist bash, zsh, Terminal, pkill or killall by name. Scope exceptions to a verified management workflow and retain burst monitoring elsewhere.

Validation Plan

Step	Action	Expected evidence
1	Run on all macOS endpoints for 30 days.	Per-device burst distribution and legitimate owners.
2	Perform a controlled kill-loop test.	Measured capture rate for brief processes.
3	Review LaunchAgent and parent evidence.	Ability to separate malware-like scripts from MDM.
4	Tune threshold only from data.	Documented percentile and expected alert volume.

Decision	Criterion
Promote / retain	No recurring benign bursts or reliable separation through parent/script ownership.
Tune	Adjust targets or threshold using measured macOS telemetry, not intuition.
Reject	Process sampling makes the signal unreliable or authorized automation is indistinguishable.

OPERATING NOTE

Confidence is Medium because the behavior is well supported but the EDR capture rate and alert threshold remain unvalidated.

ATT&CK; Mapping & References

Technique	Relationship	Mapping posture
T1562.001 - Impair Defenses	Repeated termination disrupts visible components and notifications.	Behavioral relationship.
T1056.002 - GUI Input Capture	Credential prompts are used to obtain user input.	Campaign relationship.
T1543.001 - Launch Agent	LaunchAgents provide persistence in the analyzed chain.	Supporting context.

Primary and authoritative references

- 1. Group-IB - ClickLock Stealer: Paste Once, Lose Everything
<https://www.group-ib.com/blog/clicklock-stealer-macos-malware/>
- 2. Microsoft Learn - DeviceProcessEvents schema
<https://learn.microsoft.com/en-us/defender-xdr/advanced-hunting-deviceprocessevents-table>

DOCUMENT STATUS

Research-derived and unvalidated in the target environment. The package is suitable for controlled testing and analyst review; it is not evidence that a specific environment is compromised.